Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions.

Dataset Overview

3,900	18	$59.76	37
Transactions	Features	Avg Purchase	Missing Values
Purchase records analyzed	Data points per transaction	USD per transaction	Review ratings imputed

Key features include customer demographics, purchase details, and shopping behavior metrics.

Python Data Preparation

Data Loading & Exploration

Imported dataset using `pandas` and checked structure with `df.info()` and `.describe()`.

Cleaning Process

Handled missing values in `Review Rating` using median ratings by product category.

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Column Standardization

Rewrote columns to snake case for better readability.

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Feature Engineering

Created `age_group` and `purchase_frequency_days` columns.

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Database Integration

Connected to PostgreSQL and loaded cleaned DataFrame.

SQL Analysis: Revenue Insights

Gender Revenue

Male: $157,890 | Female: $75,191

Age Groups

Young Adult: $62,143 | Middle-aged: $59,197 | Adult: $55,978 | Senior: $55,763

Shipping Types

Standard: $58.46 avg | Express: $60.48 avg

Product Performance

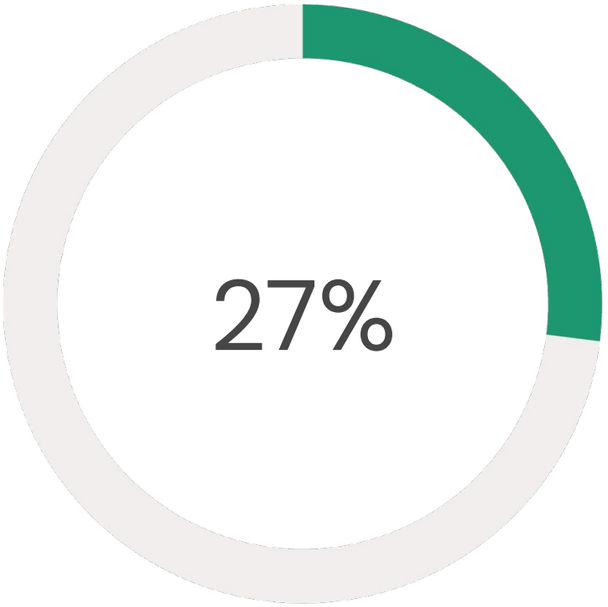
Top-Rated Products

1. Gloves (3.86)
2. Sandals (3.84)
3. Boots (3.82)
4. Hat (3.80)
5. Skirt (3.78)

Most Ordered Products

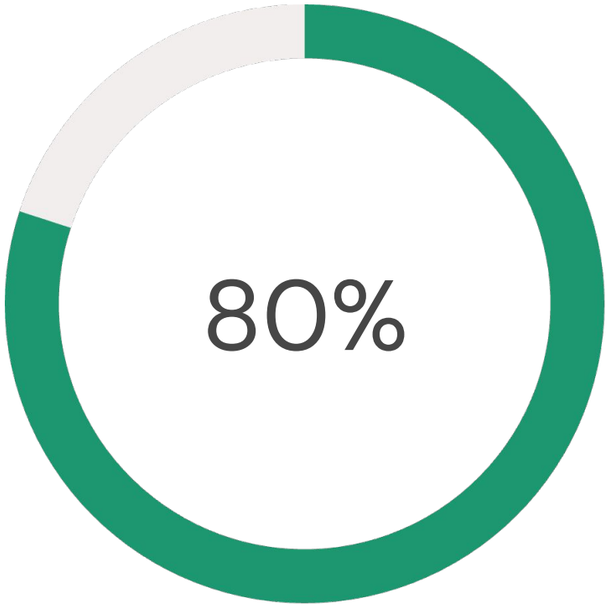
- Blouse: 171 orders
- Jewelry: 171 orders
- Pants: 171 orders
- Jacket: 163 orders

Subscription & Customer Behavior



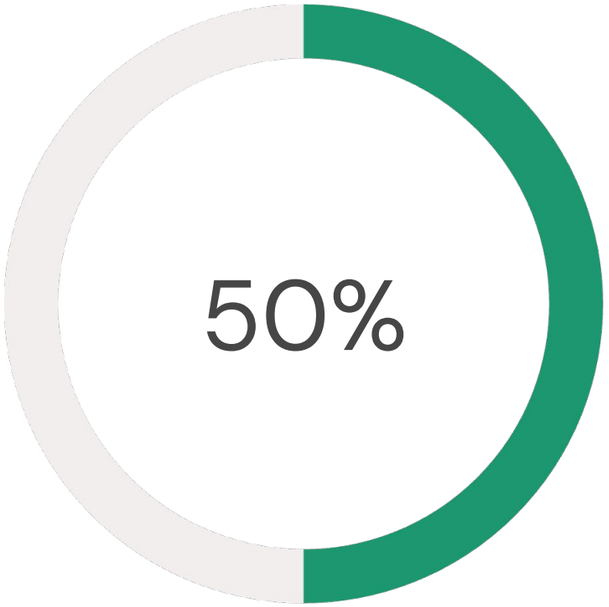
Subscribers

1,053 customers subscribed



Loyal Customers

3,116 customers in loyal segment



Discount Rate

Hats most frequently discounted

Non-subscribers spent slightly more ($59.87 vs $59.49), but subscribers showed higher engagement.

Customer Segmentation

New

83 customers with first purchase

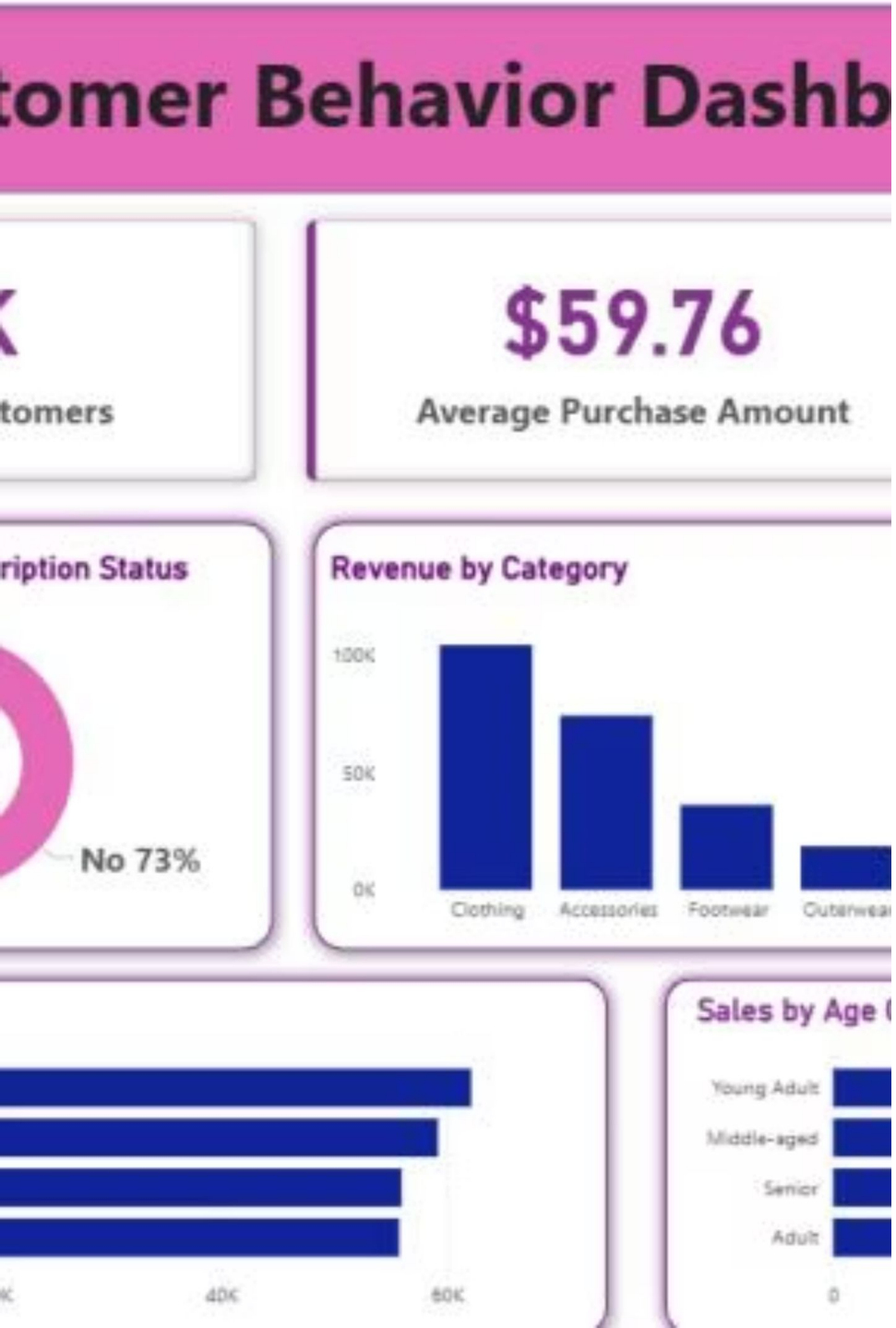
Returning

701 customers with 2-5 purchases

Loyal

3,116 customers with >5 purchases

Repeat buyers (958) showed higher subscription rates, indicating loyalty drives subscription.



Power BI Dashboard

KPIs

3.9K customers, $59.76 avg purchase, 3.75 avg rating

Filters

Subscription, gender, category, shipping type

Visualizations

Donut charts, bar charts, horizontal bars

Key Findings

Gender Gap

Male customers generated 2.1x more revenue than female customers.

Discount Strategy

50% of hat purchases used discounts, highest among all products.

Customer Loyalty

80% of customers are loyal with more than 5 purchases.

Age Impact

Young adults generated highest revenue ($62,143), seniors lowest ($55,763).

Strategic Recommendations

Boost Subscriptions

Promote exclusive benefits for subscribers to increase conversion from 27%.

Loyalty Programs

Reward repeat buyers to accelerate movement into loyal segment.

Review Discount Policy

Balance sales boosts with margin control, especially for high-discount products.

Targeted Marketing

Focus on high-revenue age groups and express-shipping users.

Product Positioning

Highlight top-rated and best-selling products in campaigns.